A HYBRID GRID-BASED METHOD FOR VIDEO REPRESENTATION

Miaoran Zhao, Mufan Liu, Wenjie Huang, Puyue Hou, Yiling Xu[*]

Cooperative Medianet Innovation Center, Shanghai Jiao Tong University, Shanghai, China
{zmr-ovo, sudo_evan, huangwenjie2023, houpy2016, yl.xu}@sjtu.edu.cn

ABSTRACT

Implicit neural representations (INRs) have shown great potential for video representation. However, existing methods rely on either positional index embeddings or content-aware embeddings as inputs to neural networks for frame reconstruction. These designs yield overly implicit representations that are difficult to compress, while the dependence on complex networks further slows inference. In this paper, we present a hybrid video representation that combines explicit multi-resolution grids with a lightweight neural network. The grids form a structural-aligned latent space that provides strong coherence for compression. They also offload most of the representational burden, allowing a smaller neural network for faster inference. Adaptive positional encoding with learnable frequencies enriches frequency-aware information and enhances the representational capacity of grid latent features. To effectively fuse grid features and positional embeddings, a gating mechanism performs dynamic fusion depend on the content, while a time-aware modulation module is introduced to strengthen the modeling of temporal variations. For compression, an entropy model is applied to the grid features and quantization to the neural network, jointly improving reconstruction quality and compression efficiency. Our method outperforms existing INR approaches, achieving up to a 5.82% PSNR gain and a $3.0\times$ to $5.4\times$ inference speedup, while improving rate-distortion performance for video compression.

***Index Terms*—** Implicit neural representation, video representation, inference speed, end-to-end compression

1. INTRODUCTION

Driven by the rapid progress of deep learning, video representation has made significant advances in recent years. Among emerging approaches, implicit neural representations (INRs), which model videos directly with neural networks, provide a powerful framework for tasks such as compression and inpainting. Broadly, INRs can be categorized into index-based and hybrid methods. Index-based methods, pioneered by NeRV [1], represent videos as mappings from frame indices to image frames. Subsequent works such as E-NeRV [2] introduce disentangled spatiotemporal context while HiNeRV [3] employs lightweight layers with hierarchical positional encoding. Despite these advances, index-based methods still rely on content-agnostic embeddings and cannot adapt to video content. To address this limitation, hybrid methods introduce explicit embeddings for content-adaptive modeling. For example, CNeRV [4] and HNeRV [5] use content-aware embeddings to enhance representational capacity and FFNeRV [6] integrates optical flow to better exploit temporal redundancy.

In spite of their success in modeling high-quality video content, INR-based approaches still exhibit several limitations. Index-based methods [1–3] depend on fully-implicit neural networks to learn content-agnostic mappings, which often lack prior knowledge and result in blurred reconstructions of high-frequency details. In addition, their architectural complexity incurs substantial computational overhead, leading to relatively slow inference and convergence. Hybrid methods [4–8] alleviate these issues to some extent, but their explicit embeddings provide only limited information, leaving most content to be captured by complex decoders. This design remains suboptimal in terms of representational efficiency and inference speed. Moreover, both categories of methods typically employ model quantization [9–12] with fixed step sizes. This strategy fails to adapt to the actual probability distribution of the parameters, which limits compression flexibility. Motivated by the advantages of feature grids in 3D visual fields [13–16], we introduce a hybrid video representation framework based on multi-resolution grids, aiming to accelerate inference, improve compression efficiency, and preserve high-fidelity reconstruction.

In the proposed framework, video is represented explicitly by multi-resolution grids and implicitly by a lightweight neural network. The grids provide coordinate-dependent latent features through trilinear interpolation, which capture spatiotemporal structures and content details. To enhance representational capacity, adaptive positional embeddings are incorporated into the latent features, where frequency parameters are optimized during training to adjust frequency levels toward more effective ranges. A gating mechanism then fuses the grid features with adaptive positional embeddings, allowing the network to balance complementary spatial-frequency information. The fused features are further processed by a time-aware modulation module, which applies channel-wise modulation conditioned on the time index to better capture temporal variation. Finally, the modulated features are decoded by a lightweight multilayer perceptron (MLP) to reconstruct video frames. For compression, an entropy model [17] is applied to enable end-to-end rate-distortion optimization of the grids, while quantization is applied to the neural network weights.

In summary, our contributions are as follows:

- We propose a hybrid video representation framework, which achieves efficient inference and high-quality reconstruction within a compact architecture.
- We employ an entropy model for end-to-end compression of grid features and apply quantization to neural network weights, improving overall compression efficiency.
- Extensive experiments demonstrate that our method outperforms existing INR approaches, achieving up to a 5.82% PSNR gain and a $3.0\times$ to $5.4\times$ inference speedup.

This paper is supported in part by National Natural Science Foundation of China (62371290), National Key R&D Program of China (2024YFB2907204), STCSM under Grant (24511107100), Shanghai Key Lab of Digital Media Processing and Transmission, Shanghai Jiao Tong University. The corresponding author is Yiling Xu (e-mail: yl.xu@sjtu.edu.cn).

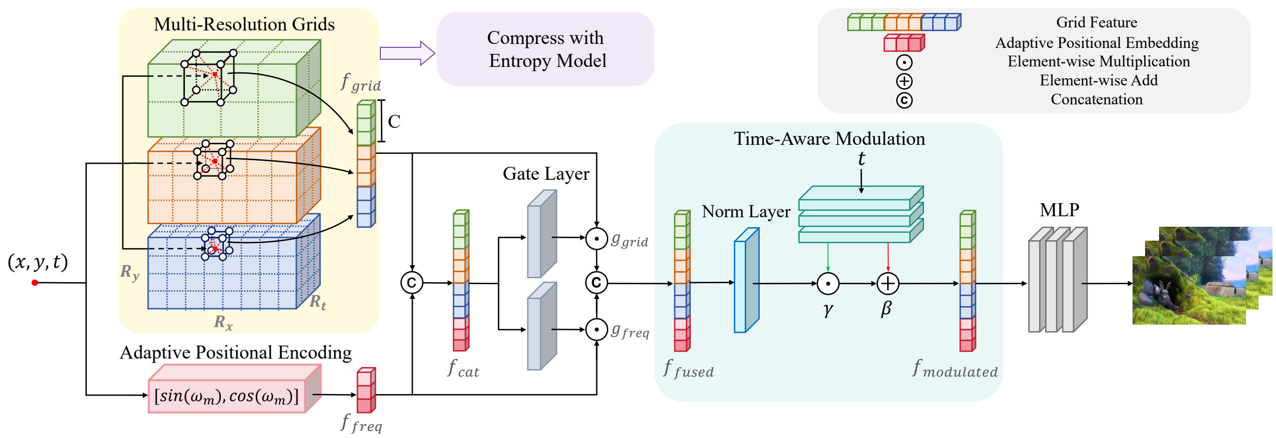

Fig. 1. Overview of the proposed video representation framework. Multi-resolution grids interpolate latent features, which are concatenated with adaptive positional embeddings. The concatenated features are refined by a gating mechanism and time-aware modulation, and then decoded by a lightweight MLP. For compression, an entropy model is applied to the grid features and quantization to the neural network.

2. METHODS

2.1. Overview

Fig. 1 illustrates the overall architecture of the proposed video representation framework. The process begins with multi-resolution grids, which interpolate latent features based on spatiotemporal coordinates. The grid features are concatenated with adaptive positional embeddings to capture information across different frequency levels. To improve temporal modeling, the concatenated features are processed through a gating mechanism and further modulated by a time-aware module. Finally, a lightweight MLP is employed to decode the modulated features and reconstruct the video frames. An entropy model [17] is utilized for end-to-end compression of the feature grids, and quantization is applied to the remaining neural network weights.

2.2. Feature Encoding

Multi-Resolution Grids: Unlike index-based representations, our framework employs explicit multi-resolution grids to encode video latent features. A single grid level is parameterized as a learnable tensor $G \in \mathbb{R}^{R_x \times R_y \times R_t \times C}$, where R_x, R_y, and R_t denote the horizontal spatial, vertical spatial, and temporal resolutions, respectively, and C is the feature dimension. The spatial resolutions R_x and R_y are scaled according to the video aspect ratio $\alpha = (\alpha_x, \alpha_y)$, while the temporal resolution R_t is adjusted independently, increasing proportionally with the number of video frames and the dynamic variations between frames:

$$\begin{aligned} R_x &= \max\left(R_{\min}, \left\lfloor R_l \cdot \frac{\alpha_x}{\min(\alpha)} \right\rfloor\right), \\ R_y &= \max\left(R_{\min}, \left\lfloor R_l \cdot \frac{\alpha_y}{\min(\alpha)} \right\rfloor\right), \\ R_t &= \max\left(R_{\min}, \lfloor R_l \cdot \beta \rfloor\right), \end{aligned} \tag{1}$$

where $R_{\min}$ is the lower bound of the resolution, R_l is the base resolution of the current level, β is the temporal scaling factor, and $\lfloor \cdot \rfloor$ denotes the floor operation. This resolution setting aligns with the video aspect ratio and temporal dimension, ensuring spatiotemporal consistency between the grid features and the video content, thereby improving reconstruction quality.

Multi-resolution grids are constructed to capture information ranging from coarse structures to fine-grained details across different levels. Let R_c and R_f denote the coarsest and finest resolutions, and L the total number of grid levels. The base resolution of the l-th level is defined as

$$R_l = R_c \cdot b^l, \quad b = \exp\left(\frac{\log(R_f) - \log(R_c)}{L - 1}\right). \tag{2}$$

Given a normalized coordinate (x, y, t), the grid feature is obtained by trilinear interpolation from its eight neighboring vertices:

$$f_{\text{grid}}(x, y, t) = \sum_{i=0}^{7} w_i \cdot G_{x_i, y_i, t_i}, \tag{3}$$

where x_i, y_i, t_i denotes the coordinate of a neighboring grid vertex, and w_i is its corresponding interpolation weight. This lookup can be efficiently implemented in parallel on GPUs, thereby improving inference speed.

Adaptive Positional Encoding: We employ adaptive positional encoding to map normalized spatial-temporal coordinates (x, y, t) into a learnable frequency space. The encoding is defined as

$$\begin{aligned} f_{\text{freq}}(x, y, t) &= \text{flatten}\left([\sin(\omega_m v), \cos(\omega_m v)]_{v \in \{x, y, t\}}\right), \\ \omega_m &= e^m, m = 0, 1, \ldots, M - 1 \end{aligned} \tag{4}$$

where M denotes the number of frequency levels. The frequency parameters ω_m are initialized exponentially, which provides a structured frequency prior to capture both low- and high-frequency information at the start of training. Unlike fixed encoding, these parameters are further optimized to adaptively refine the frequency components, yielding more expressive representations.

2.3. Feature Processing

Gating Mechanism: In video representation, the relative importance of grid features f_{grid} and adaptive positional embeddings f_{freq} varies across different content. To adaptively balance their contributions, we introduce two lightweight gating units. Given the concatenated input $f_{\text{cat}} = [f_{\text{grid}}, f_{\text{freq}}]$, each gating unit applies a linear projection followed by a sigmoid activation to produce channel-wise weights:

$$g_{\text{grid}} = \sigma(W_1 f_{\text{cat}} + b_1), \quad g_{\text{freq}} = \sigma(W_2 f_{\text{cat}} + b_2), \tag{5}$$

where W_1, W_2, b_1, and b_2 are learnable parameters, and $\sigma(\cdot)$ denotes the sigmoid function. The weights g_{grid} and g_{freq} dynamically rescale their corresponding features, leading to the fused representation:

$$f_{\text{fused}} = [g_{\text{grid}} \odot f_{\text{grid}}, g_{\text{freq}} \odot f_{\text{freq}}], \tag{6}$$

$\odot$ denoting element-wise multiplication. This gating design allows the network to selectively emphasize either spatial-temporal grid information or frequency-aware positional embeddings depending on the content.

Time-Aware Modulation: To further enhance the modeling of temporal dynamics, we modulate the fused feature using channel-wise affine transformations conditioned on the time index t. Given f_{fused} and normalized input t, a lightweight MLP produces scaling factors $\gamma(t)$ and offsets $\beta(t)$. The modulation is defined as

$$f_{\text{modulated}} = \mathcal{N}(f_{\text{fused}}) \odot \gamma(t) + \beta(t), \tag{7}$$

where $\mathcal{N}(\cdot)$ denotes channel-wise normalization:

$$\mathcal{N}(f_{b,c,h,w}) = \frac{f_{b,c,h,w} - \mu_{b,c}}{\sqrt{\sigma_{b,c}^2 + \epsilon}}, \tag{8}$$

b, c, h, and w indexing the batch, channel, height, and width of the feature, respectively. $\mu_{b,c}$ and $\sigma_{b,c}^2$ are the mean and variance of channel c over spatial dimensions. Normalization of feature prior to modulation effectively eliminates inter-channel scale discrepancies, thereby facilitating stable convergence and improving modulation performance.

2.4. End-to-end Compression

Unlike index-based methods that rely solely on neural networks, most of the information in our framework is stored in feature grids, which exhibit strong internal correlations. This property makes them fundamentally different from network parameters typically handled by pruning or quantization, and allows end-to-end compression optimization to be performed at larger scales.

Since multi-resolution grids capture distinct information at different levels, their probability distributions and sensitivity to quantization also vary. To accommodate this, an entropy model [17] is applied at each level for independent probabilistic modeling and bit estimation, thereby reducing redundancy and improving compression efficiency. The grid-based latent features f are first quantized into discrete symbols $\hat{f}$. Since the quantization operator is non-differentiable, it is approximated by adding uniform noise:

$$\hat{f} \approx f + \mathcal{U}\left(-0.5, 0.5\right). \tag{9}$$

The quantized features $\hat{f}$ are then modeled by the entropy model that learns the probability distribution $p_\theta(\hat{f})$, where θ are the learnable parameters. Based on information-theoretic principles, the bits for encoding the features are defined as the expected negative log-likelihood under this distribution:

$$R \approx \mathbb{E}_{\hat{f} \sim p_\theta}\left[-\log_2 p_\theta(\hat{f})\right]. \tag{10}$$

During training, the bit rate R serves as a regularization term, combined with the reconstruction distortion D to constitute the rate-distortion joint loss function:

$$\mathcal{L} = \lambda R + D(x, \hat{x}), \tag{11}$$

where λ controls the trade-off between bit rate and distortion and D is the L2 loss, i.e., the mean squared error (MSE) between the ground truth image x and the reconstructed image $\hat{x}$.

Table 1. Video representation results (PSNR/dB) on Bunny

Size	0.35M	0.75M	1.5M	3M	avg.↑
NeRV	27.50	30.39	32.34	33.50	30.93
E-NeRV	28.02	31.18	34.27	37.31	32.70
FFNeRV	28.56	31.45	34.68	37.46	33.04
HNeRV	27.81	32.18	35.05	37.59	33.16
Ours	**31.43**	**34.01**	**35.85**	**37.88**	**34.79**

Table 2. Model inference speed (FPS) on DAVIS

Sequence	bmx	boat	camel	cows	dance	swan	avg.↑
NeRV	84.69	84.39	83.92	84.27	83.49	83.16	83.99
E-NeRV	53.28	52.58	54.22	52.86	51.92	54.35	53.20
FFNeRV	151.49	153.52	156.20	154.98	157.43	147.84	153.58
HNeRV	62.71	62.80	62.83	64.65	64.47	64.02	63.58
Ours	**220.86**	**222.11**	**221.88**	**220.07**	**232.12**	**241.01**	**226.34**

3. EXPERIMENTS

3.1. Datasets and Implementation Details

We evaluate our method on three public datasets: Big Buck Bunny (Bunny), DAVIS [18] and UVG [19]. Bunny is sourced from scikit-video and consists of 132 frames at 720p resolution. For DAVIS, we select six representative video sequences and resize all frames to 720p. UVG comprises seven 1080p sequences at 120 FPS of 5s or 2.5s, and evaluation is conducted on consecutive 120-frame segments of each sequence.

We compare our method with four representative video INRs: index-based methods NeRV and E-NeRV, and hybrid methods FFNeRV and HNeRV. In the original configuration of HNeRV, 720p/1080p inputs are cropped to $640 \times 1280/960 \times 1920$. To ensure a fair comparison, we adjust its strides to fit the resolutions of $720 \times 1280/1080 \times 1920$. Our model is optimized using AdamW [20] with a learning rate of 5×10^{-3} and a cosine-annealing schedule [21].

For video reconstruction, all methods are trained for 1200 epochs to ensure full convergence and evaluated using peak signal-to-noise ratio (PSNR) [22] and multi-scale structural similarity index (MS-SSIM) [23]. Inference speed is reported as frames per second (FPS) and measured every 50 epochs by averaging the runtime of 10 consecutive runs. For compression evaluation, we set the number of training epochs to 300 and use rate-distortion curve to compare compression efficiency.

3.2. Main Results

We first compare our method with baselines on Bunny, as shown in Table 1. Our method achieves the highest reconstruction quality across all four model sizes. Notably, it maintains strong performance even at a small scale (0.35M parameters). It improves average PSNR by 5.82% and MS-SSIM by 2.21% on this dataset. We also evaluate all methods at the 3.0M model size on the DAVIS and UVG benchmarks; although the tables are omitted due to space, our method still outperforms the baselines, with PSNR gains of 4.93% and 2.74% and MS-SSIM improvements of 1.97% and 1.22%, respectively. Fig. 2 shows visual results on the *bmx* sequence from DAVIS, where at comparable size (3.0M) our method preserves finer details and produces reconstructions closer to the ground truth.

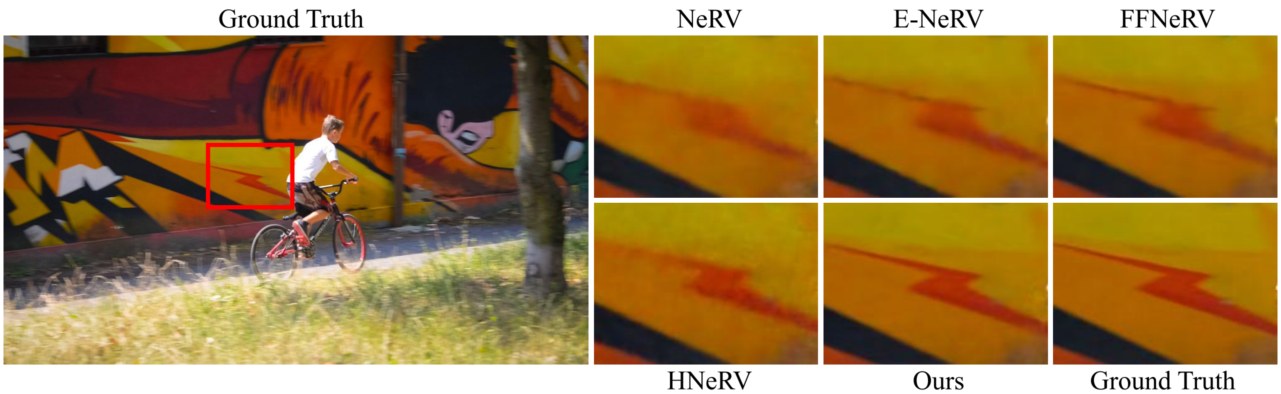

Fig. 2. Visual comparison of video regression.

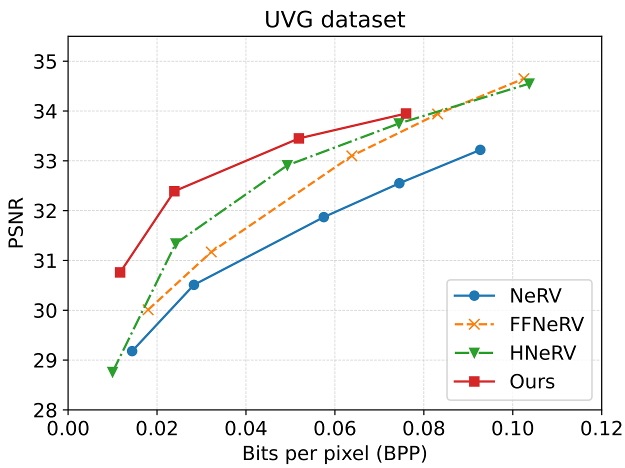

Fig. 3. Video compression results on UVG.

3.3. Model Inference

We evaluate the inference speed of all methods across three datasets under the same experimental settings. Table 2 shows the results on DAVIS with 3.0M model size, where our method delivers the highest FPS among the baselines. On average, it achieves $3.0\times$ higher FPS on DAVIS (3.0M), $2.2\times$ on Bunny (across four model sizes), and $5.4\times$ on UVG (3.0M). These results indicate that the proposed framework enables a more efficient reconstruction pipeline.

3.4. Video Compression

We train our models at different sizes using a dynamic λ strategy. During the warm-up phase, λ is set to 0 to prioritize reconstruction quality. After warm-up, λ increases linearly to its maximum value $\lambda_{\max}(5 \times 10^{-4})$, enabling the grids to progressively learn the trade-off between distortion and rate. This strategy achieves high compression efficiency across videos with diverse content while preserving reconstruction fidelity. It also prevents instabilities and significant PSNR degradation caused by large λ values during early training stages. For baselines, we follow their official implementations with pruning, 8-bit quantization and entropy encoding.

Fig. 3 shows the overall rate-distortion curves on UVG, in which our model outperforms the baselines. At the same bpp, our method yields a maximum PSNR improvement of 7.25%. It indicates that dynamically allocating bits is more effective than applying a fixed bit to all network parameters across all videos.

Table 3. Ablation study on Bunny with 3.0M size

Grids	APE	Gate	TAM	PSNR (dB)	Gain (%)
✓				37.18	0.00
✓	✓			37.25	+0.19
✓	✓	✓		37.42	+0.65
✓	✓	✓	✓	**37.88**	**+1.88**

3.5. Ablation Study

To assess the contribution of each component in the proposed framework, we conduct ablation experiments on Bunny with the 3.0M model size, as summarized in Table 3, where Grids denotes multi-resolution grids, APE denotes adaptive positional encoding, Gate denotes the gating mechanism, and TAM denotes time-aware modulation. Using only Grids yields a PSNR of 37.18 dB, demonstrating their ability to capture spatiotemporal structures and fine details. Adding APE provides a modest improvement, confirming the benefit of frequency-aware information. Moreover, incorporating Gate further raises PSNR to 37.39 dB, showing that fusing grid features with positional embeddings enhances feature complementarity. Finally, the inclusion of TAM improves PSNR to 37.88 dB, underscoring its role in modeling temporal dynamics. Overall, the results demonstrate that each component contributes positively to the final performance and validate the design choices of the proposed framework.

4. CONCLUSION

In this paper, we propose a hybrid video representation framework that combines explicit multi-resolution grids with a lightweight neural network. Our method achieves efficient inference and high-fidelity reconstruction within a compact architecture. In addition, end-to-end compression with entropy model and network quantization improve the rate-distortion performance. Extensive experiments demonstrate the consistent advantages of our method in reconstruction quality, inference speed, and compression efficiency over existing INR approaches. In the future, we will further explore the optimization of model structures and end-to-end compression based on entropy model.

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